DIGITAL THERMOMETER USING ARDUINO AND LCD

PROJECT REPORT

Submitted By

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Domain : Embedded Systems

Project Title : Digital Thermometer with LCD

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I express my sincere gratitude to the Department of Electronics and Communication Engineering, Parul University, for providing me with the opportunity to carry out this project titled "Digital Thermometer Using Arduino and LCD".

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ABSTRACT

Temperature measurement is an essential requirement in various industrial, medical, and domestic applications. This project presents a digital thermometer using an Arduino Uno, TMP36 temperature sensor, and a 16×2 LCD display. The temperature sensor continuously senses the surrounding temperature and converts it into an analog voltage signal.

The Arduino processes this signal, calculates the temperature in degree Celsius, and displays the result on the LCD. The system provides accurate and real-time temperature monitoring. The project is simple, cost-effective, and suitable for educational purposes and practical applications.

Keywords: Arduino Uno, TMP36 Sensor, LCD Display, Temperature Monitoring, Embedded Systems.

CHAPTER 1: INTRODUCTION

A digital thermometer is an electronic device used to measure temperature and display it in a digital format. Traditional thermometers use mercury or alcohol, whereas digital thermometers utilize electronic sensors for measurement.

In this project, a TMP36 temperature sensor is used to detect ambient temperature. The sensor output is connected to the Arduino Uno microcontroller, which processes the sensor data and displays the measured temperature on a 16×2 LCD screen.

This project demonstrates the integration of sensors, microcontrollers, and display modules in embedded systems.

CHAPTER 2: OBJECTIVES

The main objectives of this project are:

1. To measure ambient temperature electronically.
2. To display temperature on a digital LCD.
3. To learn Arduino programming.
4. To interface sensors with microcontrollers.
5. To understand analog-to-digital conversion.
6. To develop a low-cost temperature monitoring system.

CHAPTER 3: COMPONENTS REQUIRED

Component	Quantity
Arduino Uno	1
TMP36 Temperature Sensor	1
LCD 16×2	1
Potentiometer 10kΩ	1
Breadboard	1
Jumper Wires	As Required
USB Cable	1

CHAPTER 4: HARDWARE DESCRIPTION

Arduino Uno

Arduino Uno is based on the ATmega328P microcontroller. It contains 14 digital I/O pins and 6 analog input pins.

Features

- Operating Voltage: 5V
- Digital I/O Pins: 14
- Analog Inputs: 6
- Flash Memory: 32 KB
- Clock Speed: 16 MHz

TMP36 Temperature Sensor

The TMP36 is a low-voltage precision temperature sensor.

Features

- Temperature Range: -40°C to 125°C
- Supply Voltage: 2.7V to 5.5V
- Linear Output
- High Accuracy

LCD 16×2 Display

The LCD module displays 16 characters in 2 rows. It is used to show the measured temperature.

CHAPTER 5: SOFTWARE DESCRIPTION

Tinkercad

Tinkercad is an online simulation platform used for designing and testing electronic circuits.

Arduino IDE

Arduino IDE is used for writing, compiling, and uploading Arduino programs.

Programming Language:

- Embedded C/C++

CHAPTER 6: CIRCUIT DIAGRAM

Connections

TMP36 Sensor:

- VCC → 5V
- Output → A0
- GND → GND

LCD:

- VSS → GND
- VDD → 5V
- V0 → Potentiometer
- RS → Pin 12
- EN → Pin 11
- D4 → Pin 5
- D5 → Pin 4
- D6 → Pin 3
- D7 → Pin 2
- RW → GND

Potentiometer:

- Left Pin → GND
- Middle Pin → LCD V0
- Right Pin → 5V

CHAPTER 7: WORKING PRINCIPLE

The TMP36 sensor detects ambient temperature and produces a voltage proportional to the measured temperature.

The Arduino reads the analog voltage through pin A0 using its built-in ADC. The ADC converts the analog signal into a digital value.

The Arduino calculates the temperature using:

Temperature (°C) = (Voltage − 0.5) × 100

The calculated temperature is displayed on the LCD screen.

The process repeats continuously, providing real-time temperature monitoring.

CHAPTER 8: ARDUINO PROGRAM

```
#include <LiquidCrystal.h>

LiquidCrystal lcd(12,11,5,4,3,2);

int sensorPin = A0;

void setup()
{
  lcd.begin(16,2);
}

void loop()
{
  int reading = analogRead(sensorPin);

  float voltage = reading * 5.0;
  voltage /= 1024.0;

  float temperatureC = (voltage - 0.5) * 100;

  lcd.clear();

  lcd.setCursor(0,0);
  lcd.print("Temperature");

  lcd.setCursor(0,1);
  lcd.print(temperatureC);
  lcd.print((char)223);
  lcd.print("C");

  delay(1000);
}
```

CHAPTER 9: RESULTS AND OUTPUT

The system successfully measures and displays temperature.

Sample Output:

Temperature
24.71°C

The displayed value changes automatically according to the surrounding temperature.

CHAPTER 10: APPLICATIONS

1. Room Temperature Monitoring
2. Weather Stations
3. Industrial Automation
4. Medical Equipment
5. HVAC Systems
6. Smart Home Applications
7. Laboratory Monitoring Systems

CHAPTER 11: ADVANTAGES

1. Simple and Easy to Build
2. Low Cost
3. Real-Time Monitoring
4. High Accuracy
5. Portable Design
6. Low Power Consumption

CHAPTER 12: LIMITATIONS

1. Limited Temperature Range
2. Requires External Power Supply
3. Suitable for Small-Scale Applications
4. Sensor Accuracy Depends on Calibration

CHAPTER 13: FUTURE SCOPE

1. Wireless Temperature Monitoring using IoT
2. Mobile Application Integration
3. Data Logging to Cloud Storage
4. Temperature Alerts through SMS
5. Integration with Smart Home Systems

CHAPTER 14: CONCLUSION

The Digital Thermometer Using Arduino and LCD was successfully designed and implemented. The TMP36 sensor accurately measures temperature, and the Arduino processes and displays the temperature on the LCD.

The project demonstrates the practical application of sensors, microcontrollers, and display modules in embedded systems. It is an effective educational project for learning sensor interfacing and embedded programming.